6.5 Properties of Logarithms

Name _____ Period ____ Date _____

Algebra II 6.5 (30-minute plan) · Student sheet

I can…

- use the properties of logarithms to evaluate logarithms.
- expand and condense logarithmic expressions.
- use the change-of-base formula.

Vocabulary · fill in the missing word

Product Property — $\log_b mn$ = _____

Quotient Property — $\log_b \left(\frac{m}{n}\right)$ = _____

Power Property — $\log_b m^n$ = _____

Change-of-base formula — $\log_c a$ = _____ = $\ln a / \ln c$

Part 1 · Notes and examples (fill in each note, then work the example)

Product Property: $\log_b mn$ = _____

Quotient Property: $\log_b (m/n)$ = _____

Power Property: $\log_b m^n$ = _____

Example 1

Use $\log_2 3 \approx 1.585$ and $\log_2 7 \approx 2.807$ to evaluate each logarithm.

a. $\log_2 21$ b. $\log_2 \left(\frac{7}{3}\right)$ c. $\log_2 49$

Change of base: $\log_c a$ = _____ = _____

Example 2

Evaluate $\log_3 8$ using common logarithms and natural logarithms.

Part 2 · Partner practice

Partner practice

Expand $\log_6 (3x^4)$.